

Voting Rights Act Compliance Through Edge-Weighted Graph Partitioning

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Abstract

The Voting Rights Act (VRA) mandates creation of majority-minority (MM) districts where feasible, but achieving this through principled algorithmic methods remains challenging. We systematically evaluate VRA compliance across five covered states (Alabama, Georgia, Louisiana, Mississippi, South Carolina) using four optimization approaches: recursive bisection, n-way partitioning, adaptive bisection, and edge-weighted single-objective optimization.

Multi-constraint methods (first three approaches) achieve limited success (40%, 2/5 states), with Alabama failing to reach the 50% MM threshold despite extensive testing (maximum 49.6%). We introduce edge-weighted graph partitioning, which assigns higher weights to edges between minority tracts, directly encoding the principle "don't split minority communities" into graph structure. This novel approach doubles success rate to 80% (4/5 states), achieving VRA compliance in Alabama (2 MM districts at 50.8% minority) while maintaining compactness (average +4% edge cut increase).

Through ablation study of 140 configurations (7 weight factors $\times$ 4 thresholds $\times$ 5 states), we identify optimal parameters: weight factors 5x-10x with 40-45% minority thresholds achieve VRA compliance at minimal compactness cost. Results demonstrate edge-weighting resolves the perceived tension between VRA compliance and algorithmic principles, revising feasibility thresholds from 42% (multi-constraint) to 36% state-wide minority population.

This work provides both methodological innovation (edge-weighted optimization) and practical guidance for algorithmic redistricting in VRA-covered jurisdictions.

1 Introduction

The Voting Rights Act of 1965, particularly Section 2 as amended in 1982, prohibits electoral systems that dilute minority voting strength. Courts have interpreted this to require creation of majority-minority (MM) districts where geographically possible (??). However, achieving VRA compliance through principled, algorithmic methods that avoid partisan gerrymandering remains an open challenge.

Traditional redistricting approaches face a fundamental tension: VRA compliance requires concentrating minority populations into specific districts (non-uniform distribution),

while principled redistricting emphasizes compactness and contiguity (favoring uniform distribution). Recent work on algorithmic redistricting using graph partitioning methods (???) focuses primarily on compactness and population balance, with limited attention to VRA constraints.

1.1 Research Questions

This paper investigates three fundamental questions:

1. **Feasibility:** Can principled, multi-constraint optimization methods achieve VRA compliance without resorting to partisan map-drawing?
2. **Geographic Constraints:** What demographic and geographic factors determine whether a state can achieve VRA targets through algorithmic methods?
3. **Methodological Comparison:** How do different optimization approaches (n-way partitioning, adaptive recursive bisection, constraint relaxation) perform for VRA compliance?

1.2 Key Contributions

We make three primary contributions:

(1) **Comprehensive VRA Testing:** First systematic evaluation of multi-constraint optimization for VRA compliance across multiple covered states, using METIS graph partitioning with 2D vertex weights (population + minority voting-age population).

(2) **Critical Threshold Discovery:** We identify a critical threshold around 42% state-wide minority population, above which VRA compliance is achievable with compact districts, below which it becomes infeasible without relaxing compactness.

(3) **Geographic Dominance:** We demonstrate that geographic clustering of minority populations is the primary determinant of VRA compliance feasibility, not algorithmic sophistication. Advanced methods (n-way partitioning, adaptive bisection) improve concentration by 3-7 percentage points but cannot overcome fundamental geographic constraints.

1.3 Research Program Context

This paper extends the foundational recursive bisection method presented in ?, which establishes algorithmic objectivity through structural immunity to partisan manipulation, and builds on the edge-weighted approach in ?, which optimizes geometric compactness. While those baseline methods achieve population balance and compactness, they do not explicitly address Voting Rights Act requirements for majority-minority district creation. This work adds demographic awareness to the partitioning framework, investigating whether principled algorithmic methods can satisfy VRA obligations while maintaining the impossibility defense against gerrymandering. Subsequent papers in this research program investigate the demographic thresholds that determine VRA feasibility and the compactness tradeoffs inherent in creating majority-minority districts.

1.4 Overview

Section 2 reviews VRA requirements and prior algorithmic redistricting work. Section 3 describes our multi-constraint optimization methodology and four tested approaches. Section 4 presents results across five VRA-covered states. Section 5 discusses implications for redistricting practice and policy. Section 6 concludes with recommendations for future work.

2 Background and Related Work

2.1 Voting Rights Act Requirements

Section 2 of the Voting Rights Act, as amended in 1982, prohibits any voting practice that “results in a denial or abridgement of the right to vote on account of race or color” (?). Courts apply the *Gingles* test (?) to determine whether redistricting plans violate Section 2:

1. The minority group must be sufficiently large and geographically compact to constitute a majority in a single-member district.
2. The minority group must be politically cohesive.
3. The majority votes sufficiently as a bloc to enable it to defeat the minority’s preferred candidate.

When these conditions are met, jurisdictions must create majority-minority (MM) districts where the minority voting-age population (VAP) exceeds 50%. In practice, courts often require higher thresholds (55-65%) to account for lower registration and turnout rates (?).

2.2 Algorithmic Redistricting

Recent work on algorithmic redistricting emphasizes compactness, contiguity, and population balance as core principles (???). Common approaches include:

Recursive Bisection: Hierarchically partitioning geography into districts by repeatedly splitting regions in two (?). METIS graph partitioning (?) minimizes edge cuts (compactness proxy) while balancing population.

Markov Chain Monte Carlo (MCMC): Generating ensembles of valid redistricting plans through random perturbations (?). Enables statistical analysis of partisan bias and outlier detection.

Optimization: Direct optimization of redistricting criteria using integer programming (??) or metaheuristics (?).

However, existing work focuses primarily on compactness and partisan fairness, with limited attention to VRA compliance as a hard constraint.

2.3 Multi-Constraint Graph Partitioning

METIS supports multi-constraint partitioning where vertices have multiple weights (e.g., population + minority VAP) and optimization balances all constraints simultaneously (?). Target partition weights (`tpwgts`) specify desired distribution for each constraint. Per-constraint imbalance tolerance (`ubvec`) allows relaxation of specific constraints.

Prior work uses multi-constraint partitioning for heterogeneous computing (?) and scientific computing (?), but not for VRA-aware redistricting.

2.4 VRA Compliance in Practice

Real-world redistricting combines algorithmic tools with political negotiation and legal review (?). States with VRA requirements often create “packed” MM districts with 60-70% minority VAP, sometimes at the expense of compactness (?).

Recent work (??) analyzes VRA compliance in specific states but lacks systematic comparison of algorithmic approaches or identification of feasibility thresholds.

2.5 Gap in Literature

No prior work systematically evaluates whether principled multi-constraint optimization can achieve VRA compliance across multiple covered states. Our work fills this gap by: (1) Testing four optimization approaches on five VRA states, (2) Identifying critical demographic thresholds for feasibility, (3) Quantifying the tradeoff between VRA compliance and compactness.

3 Methodology

3.1 Problem Formulation

We model redistricting as constrained graph partitioning. Census tracts form graph vertices with 2D weights $w_v = (p_v, m_v)$ representing total population and minority VAP. Edges connect geographically adjacent tracts. Goal: partition into k districts minimizing edge cuts (compactness proxy) while satisfying:

$$\text{Population balance: } \left| \frac{\sum_{v \in D_i} p_v}{\sum_v p_v} - \frac{1}{k} \right| \leq \epsilon_p \quad \forall i \quad (1)$$

$$\text{Contiguity: } D_i \text{ forms connected subgraph} \quad \forall i \quad (2)$$

$$\text{MM target: } \frac{\sum_{v \in D_i} m_v}{\sum_{v \in D_i} p_v} \geq 0.50 \quad \text{for } t \text{ districts} \quad (3)$$

where D_i is district i , $\epsilon_p = 0.005$ is population tolerance, and t is the target number of MM districts (typically 2-5 depending on state).

3.2 States and Targets

We test five VRA-covered states with varying demographics and district counts:

State	Districts	Target MM	Minority %	Tracts
Alabama (AL)	7	2	36.9%	1,437
Georgia (GA)	14	5	42.4%	2,796
Louisiana (LA)	6	2	41.6%	1,388
Mississippi (MS)	4	2	46.1%	878
South Carolina (SC)	7	3	35.1%	1,323

Table 1: VRA test states with 2020 Census demographics

Target MM district counts reflect statutory requirements and recent legal precedent. Minority VAP includes Black and Hispanic populations, following standard VRA analysis (?).

3.3 Data

Geographic: Census 2020 TIGER/Line shapefiles for tract boundaries and adjacency.

Demographic: Census 2020 redistricting files (PL 94-171) for total population and voting-age population (VAP) by race/ethnicity. *Note on denominator:* The MM district threshold (50% minority) and the minority tract identification threshold τ are applied against **total population** (not VAP) throughout this paper. Total population is the appropriate denominator for redistricting population-balance calculations under the Equal Protection Clause; VAP is the denominator courts examine for VRA dilution analysis. Where the background section references “minority VAP” in the context of Gingles preconditions, this refers to the legal standard; our algorithmic threshold uses total population as a practical proxy because PL 94-171 total population data is available at the tract level for all 50 states.

Graph Construction: Queen contiguity (shared boundary or vertex) with water-based connections for disconnected components. Final graphs: 878-2,796 vertices, average degree 6.1.

3.4 Optimization Approaches

We test four approaches combining two partitioning methods with two constraint modes:

3.4.1 Method 1: Recursive Bisection

Hierarchically split into districts by repeated binary partitioning (?). At each split of n districts into (n_L, n_R) :

1. Calculate target weights for left/right parts based on desired MM distribution
2. Run METIS 2-way partition with target weights

3. Recurse on each part until all districts formed

Tree structure matters: different binary trees yield different intermediate constraints. We test predetermined balanced trees (e.g., $7 \rightarrow [3, 4] \rightarrow \dots$).

3.4.2 Method 2: N-Way Partitioning

Partition directly into all k districts simultaneously. Specify target weights for each final district:

- MM districts ($i = 1, \dots, t$): Equal population, high minority share
- Non-MM districts ($i = t + 1, \dots, k$): Equal population, remaining minority

N-way considers all edge cuts globally, avoiding intermediate constraint limitations of recursive bisection.

3.4.3 Method 3: Adaptive Recursive Bisection

At each split, evaluate both (n_L, n_R) and (n_R, n_L) orderings. Choose whichever achieves better minority concentration. Makes locally optimal decisions rather than following predetermined tree.

3.4.4 Constraint Modes

Strict (tpwgts only): Specify target weights w_{ij} for partition i , constraint j . METIS balances both population and minority constraints simultaneously with tolerance $\epsilon = 0.005$.

Relaxed (tpwgts + ubvec): Additionally specify per-constraint imbalance: ubvec = [1.005, 1000]. Allows $1000\times$ more imbalance in minority distribution, enabling concentration while maintaining population balance.

3.5 Target Weight Calculation

For k districts with target t MM districts at 60% minority:

$$\text{MM districts: } w_{i,\text{pop}} = \frac{1}{k}, \quad w_{i,\text{min}} = \frac{0.60/k}{m_{\text{state}}} \quad i \leq t \quad (4)$$

$$\text{Non-MM: } w_{i,\text{pop}} = \frac{1}{k}, \quad w_{i,\text{min}} = \frac{(m_{\text{state}} - 0.60t/k)/k_{\text{non}}}{m_{\text{state}}} \quad i > t \quad (5)$$

where m_{state} is state-wide minority fraction and $k_{\text{non}} = k - t$ is number of non-MM districts.

3.6 Method 4: Edge-Weighted Single-Objective Optimization

Multi-constraint optimization (Methods 1-3) treats VRA compliance as a dual-objective problem, simultaneously balancing population and minority VAP. However, this approach proved insufficient for states with dispersed minority populations. We develop an alternative: **single-objective optimization with weighted edges**.

3.6.1 Edge Weighting Strategy

Rather than multi-constraint balancing, we modify the graph structure itself. For each edge $(u, v) \in E$ connecting tracts u and v :

$$w_{edge}(u, v) = \begin{cases} \alpha & \text{if } m_u \geq \tau \text{ and } m_v \geq \tau \\ 1 & \text{otherwise} \end{cases} \quad (6)$$

where m_u is the minority percentage in tract u , τ is a minority threshold (e.g., 0.45 or 0.50), and α is a weight factor (e.g., 5, 10, 100).

Key insight: Edges between high-minority tracts receive higher weights, making them "expensive" to cut. METIS minimizes total weighted edge cut, so it naturally preserves minority communities.

3.6.2 Optimization Objective

Unlike Methods 1-3 which use 2D vertex weights $w_v = (p_v, m_v)$, Method 4 uses:

- **Vertex weights:** 1D (population only): $w_v = p_v$
- **Edge weights:** As defined above
- **No target weights** (no tpwgts file)
- **No constraint relaxation** (no ubvec parameter)

This is pure single-objective edge-cut minimization with a modified graph. METIS minimizes:

$$\text{EdgeCut} = \sum_{(u,v) \in E: D_u \neq D_v} w_{edge}(u, v) \quad (7)$$

subject only to population balance constraint (??).

3.6.3 Ablation Study Design

We systematically test edge weighting across parameter space:

- **Weight factors α :** 1, 5, 10, 50, 100, 500, 1000
- **Minority thresholds τ :** 0.40, 0.45, 0.50, 0.55
- **States:** All 5 VRA-covered (AL, GA, LA, MS, SC)

Total configurations: $7 \times 4 \times 5 = 140$ tests.

For each configuration, we measure:

1. MM districts achieved (primary metric)
2. Maximum minority concentration
3. Edge cut (compactness proxy): unweighted and weighted
4. Average internal edges per district

3.6.4 Comparison to Multi-Constraint

Edge weighting differs fundamentally from Methods 1-3:

	Multi-Constraint	Edge-Weighted
Vertex weights	2D (pop + minority)	1D (pop only)
Edge weights	Uniform (1.0)	Variable (1 or α)
Optimization	Dual-objective	Single-objective
Target weights	Yes (tpwgts)	No
Constraint relaxation	Optional (ubvec)	Not applicable
Complexity	High	Low

Table 2: Multi-constraint vs edge-weighted optimization

Hypothesis: Edge weighting directly encodes the VRA goal (“don’t split minority communities”) into graph structure, potentially achieving better results than indirect multi-constraint balancing.

3.7 Production Algorithm: Adaptive Weight Formula

The ablation study above tests fixed weight factors to characterize the parameter space. The deployed V4 pipeline uses an **adaptive formula** that sets α automatically based on each state’s minority tract density:

$$\alpha = \max(3.0, 10.0 \times (1.0 - 0.7 \times f_{\text{minority}})) \quad (8)$$

where f_{minority} is the fraction of tracts in the state with minority percentage $\geq \tau$ (the minority threshold, default $\tau = 0.40$).

Rationale: States with densely clustered minority tracts (high f_{minority}) need less boosting—the geography already concentrates minority tracts together. States with sparse minority tracts (low f_{minority}) need stronger signals to overcome dispersion. The formula scales from $10\times$ (no minority tracts, $f = 0$) down to a floor of $3\times$ (saturated states), with the 0.7 coefficient tuned to keep α in the empirically effective $5\times$ – $10\times$ range for the covered Southern states.

Effective weights for key states (2020 Census, $\tau = 0.40$):

State	f_{minority}	α (adaptive)	Closest fixed config
Alabama	~ 0.22	$\approx 8.5\times$	$10\times$
Georgia	~ 0.34	$\approx 7.6\times$	$10\times$
Louisiana	~ 0.30	$\approx 7.9\times$	$10\times$
Mississippi	~ 0.47	$\approx 6.7\times$	$5\times$
South Carolina	~ 0.16	$\approx 8.9\times$	$10\times$
California	~ 0.63	$\approx 5.6\times$	$5\times$

Relationship to ablation: The adaptive formula maps each state to an effective α within the $5\times$ – $10\times$ range that the ablation identifies as optimal (Section ??). Results for

the ablation’s fixed-weight configurations are therefore directly comparable to V4 production outputs for all covered states except those near the $3\times$ floor (high-diversity states such as California, Hawaii).

V4 production outcomes (verified 2026-04-24, current pipeline):

State	Target MM	V4 MM Achieved	MM District Minority %
Alabama	2	2	51.1%, 50.5%
Georgia	5	7	52.3%–71.4%
Louisiana	2	1	60.2%
Mississippi	2	1	55.8% (D1 at 49.99% [†])
South Carolina	3	1	52.6%

[†]Mississippi District 1 reaches 49.99% minority—a stochastic boundary condition. Across repeated METIS runs with different random seeds, this district fluctuates across the 50% threshold. The adaptive formula’s effective $\alpha \approx 6.7\times$ is between the ablation’s $5\times$ and $10\times$ configs; achieving both MM districts reproducibly would require the $10\times$ – $50\times$ range or a lower threshold.

Discrepancy from ablation headline: The ablation study reports 80% success (4/5 states) using per-state-optimal fixed configurations. The V4 production adaptive formula achieves 40% of full legal targets (Alabama and Georgia only), with Louisiana and Mississippi each at 1/2 districts. Section ?? discusses this gap.

3.8 Implementation

Partitioner: METIS 5.1.0 (`gpmets`) with flags `-contig` (enforce contiguity), `-minconn` (minimize subdomain connectivity), `-niter=100` (refinement iterations).

Platform: Python 3.13 with GeoPandas for spatial operations, NumPy for weight calculations.

Analysis: Count MM districts with $\geq 50\%$ minority population. Report maximum minority concentration achieved and number of districts meeting threshold. For edge-weighted approach, additionally measure edge cut (unweighted and weighted) to assess compactness tradeoff.

All code and data available at [repository URL upon acceptance].

4 Results

4.1 Overall Success Rates

Table ?? shows VRA compliance across all five states and four methods.

Key Finding 1: Only 2 of 5 states (40%) achieve VRA targets with principled methods. Georgia and Mississippi consistently succeed across all approaches. Alabama, South Carolina, and Louisiana (partial) fail to meet targets.

State	Target MM	N-Way		Adaptive	
		tpwgts	+ubvec	tpwgts	+ubvec
Alabama	2	0 (47.3%)	0 (49.6%)	0 (46.1%)	0 (49.1%)
Georgia	5	5 (70.4%)	5 (76.7%)	5 (est.)	5 (est.)
Louisiana	2	1 (58.8%)	1 (52.9%)	-	-
Mississippi	2	2 (53.3%)	2 (53.2%)	-	-
South Carolina	3	0 (44.4%)	0 (47.2%)	-	-
Success Rate		40% (2/5)	40% (2/5)	-	-

Table 3: MM districts achieved (maximum minority % in parentheses). Bold indicates success meeting target. Adaptive results estimated for large states.

4.2 State-by-State Analysis

4.2.1 Georgia: Complete Success

Georgia (42.4% minority, target 5 MM) achieves all 5 MM districts with both strict and relaxed constraints. N-way with tpwgts only produces 5 MM districts at 70.4% maximum minority. Adding ubvec increases concentration to 76.7% but does not change district count.

District breakdown (n-way + ubvec):

- District 7: 76.7% minority (MM)
- District 8: 74.2% minority (MM)
- District 11: 66.1% minority (MM)
- Districts 9, 13: 60-65% minority (MM)
- Remaining 9 districts: 25-45% minority

Georgia’s success reflects both high state-wide minority percentage and favorable geographic clustering in Atlanta metro, Savannah, and Southwest Georgia.

4.2.2 Mississippi: Narrow Success

Mississippi (46.1% minority, target 2 MM) achieves exactly 2 MM districts with both methods:

N-way results:

- District 2: 53.2% minority (MM)
- District 1: 52.4% minority (MM)
- District 0: 38.3% minority
- District 3: (remaining)

Both MM districts barely exceed 50% threshold. Mississippi’s 46.1% state-wide minority (highest among tested states) enables success despite smaller total population (4 districts vs. 7-14 in other states).

4.2.3 Louisiana: Partial Success

Louisiana (41.6% minority, target 2 MM) achieves only 1 MM district:

N-way + tpwgts:

- District 1: 58.8% minority (MM)
- District 0: 49.6% minority (0.4 points short)
- Remaining districts: 30-45% minority

N-way + ubvec:

- District 1: 52.9% minority (MM)
- District 2: 44.5% minority
- Remaining districts: 30-40% minority

Interestingly, adding ubvec decreases maximum concentration (58.8% $\rightarrow$ 52.9%), suggesting different local optima. Louisiana is borderline feasible - could potentially achieve 2 MM with alternative approaches or relaxed compactness.

4.2.4 Alabama: Complete Failure

Alabama (36.9% minority, target 2 MM) fails to create any MM districts across all methods:

Method	Max Minority	MM Count
Recursive [3,4] + tpwgts	43.0%	0
Recursive [4,3] + tpwgts	43.0%	0
N-way + tpwgts	47.3%	0
Adaptive + tpwgts	46.1%	0
N-way + ubvec	49.6%	0
Adaptive + ubvec	49.1%	0
Target	50.0%	2

Table 4: Alabama results across all methods. Best result (49.6%) falls 0.4 percentage points short of MM threshold.

Key observations:

- N-way outperforms recursive bisection by 4.3 percentage points (47.3% vs. 43.0%)
- Adaptive improves over predetermined trees by 3.1 points (46.1% vs. 43.0%)
- ubvec adds 2-3 percentage points but cannot cross 50% threshold
- All 6 predetermined tree structures ([6,1], [5,2], [4,3], [3,4], [2,5], [1,6]) produce identical results (43%), suggesting geography dominates tree structure

Even testing with target of only 1 MM district (instead of 2), Alabama achieves maximum 49.8% - still below threshold.

4.2.5 South Carolina: Significant Shortfall

South Carolina (35.1% minority, target 3 MM) performs worst:

N-way + ubvec (best result):

- District 0: 47.2% minority (2.8 points short)
- District 3: 42.8% minority
- Remaining districts: 30-40% minority

Like Alabama, South Carolina’s low state-wide minority percentage and dispersed geographic distribution prevent VRA compliance with compactness constraints.

4.3 Method Comparison

4.3.1 N-Way vs. Recursive Bisection

N-way partitioning consistently outperforms recursive bisection for minority concentration:

- Alabama: 47.3% vs. 43.0% (+4.3 points)
- Advantage holds with ubvec: 49.6% vs. 43.0% (+6.6 points)

However, **this does not change outcomes**: Both methods succeed on the same states (GA, MS) and fail on the same states (AL, SC). N-way produces better concentration but cannot overcome fundamental geographic constraints.

4.3.2 Adaptive vs. Predetermined Trees

Adaptive recursive bisection improves over predetermined tree structures:

- Alabama: 46.1% vs. 43.0% (+3.1 points) with tpgts
- With ubvec: 49.1% vs. 43.0% (+6.1 points)

Adaptive makes locally optimal split decisions (e.g., choosing [4,3] over [3,4] at root based on actual concentration achieved). However, like n-way, this is still insufficient to achieve VRA targets in challenging states.

4.3.3 Constraint Relaxation (ubvec)

Adding ubvec=1000 (allowing $1000\times$ imbalance in minority constraint) improves concentration in difficult states:

Key Finding 2: ubvec helps advanced methods (n-way, adaptive) more than simple recursive bisection. But it does not change which states succeed - same 2/5 success rate with or without ubvec.

State	tpwgts Only	+ubvec	Gain
Alabama (n-way)	47.3%	49.6%	+2.3
Alabama (adaptive)	46.1%	49.1%	+3.0
South Carolina	44.4%	47.2%	+2.8
Georgia	70.4%	76.7%	+6.3
Mississippi	53.3%	53.2%	-0.1

Table 5: Effect of constraint relaxation (ubvec) on maximum minority concentration

State	Minority %	Success
Mississippi	46.1%	
Georgia	42.4%	
Louisiana	41.6%	Partial (1/2)
Alabama	36.9%	
South Carolina	35.1%	

Figure 1: Success vs. state-wide minority percentage

4.4 Demographic Threshold Analysis

Plotting state-wide minority percentage vs. success reveals critical threshold:

Key Finding 3: Critical threshold around **42% state-wide minority** for VRA compliance with compactness constraints. States $\geq 42\%$ succeed, states $< 37\%$ fail, 40-42% borderline.

This threshold reflects fundamental mathematics: Creating t MM districts at 60% minority from state-wide percentage m requires:

$$m \geq \frac{0.60t}{k} + \left(1 - \frac{t}{k}\right) \cdot m_{\text{other}} \quad (9)$$

For Alabama ($k = 7$, $t = 2$, $m = 0.369$): Requires 17.1% of state as concentrated minority ($2 \times 1/7 \times 0.60$), leaving 19.8% for remaining 5 districts (39.6% each). Geographic compactness prevents this concentration.

4.5 Edge-Weighted Single-Objective Results

Multi-constraint optimization (Methods 1-3) achieved limited success: only 2 of 5 states (40%) met VRA targets. We now test whether edge-weighted single-objective optimization can overcome these limitations.

4.5.1 Ablation Study Overview

We test 140 configurations: 7 weight factors ($\alpha \in \{1, 5, 10, 50, 100, 500, 1000\}$) $\times$ 4 thresholds ($\tau \in \{0.40, 0.45, 0.50, 0.55\}$) $\times$ 5 states.

Table ?? summarizes optimal configurations per state.

State	Target MM	Multi-Constraint tpwgts +ubvec		Config	Edge-Weighted (Op MM
Mississippi	2	2 (53.3%)	2 (53.2%)	1x @ 40%	2 (53.8%) †
Georgia	5	5 (70.4%)	5 (76.7%)	1x @ 40%	5 (76.9%) ‡
Louisiana	2	1 (58.8%)	1 (52.9%)	5x @ 40%	2 (55.8%) §
Alabama	2	0 (47.3%)	0 (49.6%)	5x @ 40%	2 (50.8%)
South Carolina	3	0 (44.4%)	0 (47.2%)	50x @ 50%	0 (49.2%)
Success Rate (ablation)		40% (2/5)	40% (2/5)		80% (4/5)
Success Rate (V4 production)		40% (2/5): Alabama ✓, Georgia ✓; Louisiana 1/2, Mississippi			

Table 6: Edge-weighted vs multi-constraint VRA compliance. Bold indicates meeting target. Optimal configuration shown (lowest weight factor achieving success). V4 production row shows verified outcomes from the adaptive formula (Eq. ??).

†Mississippi District 1 reaches 49.99% in the V4 production run (2026-04-24)—essentially a stochastic boundary. The ablation’s 1x@40% configuration achieves 2 MM reproducibly; the adaptive formula’s $\approx 6.7\times$ weight is above the sufficient threshold but METIS randomness places District 1 just below 50%.

‡Georgia achieves 7 MM districts in V4 production (Districts 7,9–14; range 52.3–71.4%), exceeding the 5-district target. The ablation configuration shown achieves 5.

§Louisiana achieves only 1 MM (60.2%) in V4 production. The ablation’s 5x@40% achieves 2 (55.8%). See Section ?? for analysis of this production gap.

Key Finding 4: Edge-weighted optimization achieves 80% success rate (4/5 states) with per-state-optimal fixed configurations, doubling multi-constraint performance (40%). With the V4 production adaptive formula, full legal targets are met in 2/5 states (Alabama, Georgia), with Louisiana and Mississippi each achieving 1 of 2 target districts—see Section ??.

4.5.2 Alabama: Breakthrough Result

Alabama represents the critical case where multi-constraint failed (maximum 49.6% minority, 0.4 points short). Edge weighting succeeds:

Multiple edge-weighted configurations achieve 2 MM districts. The most principled (lowest weight factor) is 5x @ 40%, creating districts at 50.8% and 50.1% minority while maintaining edge cut of 276 (baseline: 280).

Key Finding 5: Edge weighting crosses Alabama’s 50% threshold, achieving VRA compliance where multi-constraint fundamentally failed.

4.5.3 Louisiana: Partial to Full Success

Louisiana achieved only 1 of 2 target MM districts with multi-constraint (58.8% max). Edge weighting achieves both:

Optimal configuration (5x @ 40%):

- District 1: 55.8% minority (MM)

Configuration	MM Districts	Max Minority
<i>Multi-Constraint (Methods 1-3)</i>		
N-way + tpwgts	0	47.3%
N-way + ubvec	0	49.6%
Adaptive + ubvec	0	49.1%
<i>Edge-Weighted (Method 4)</i>		
Baseline (1x @ 50%)	0	49.0%
5x @ 40%	2	50.8%
50x @ 40%	2	51.8%
10x @ 45%	2	52.0%
100x @ 45%	2	51.4%

Table 7: Alabama results: multi-constraint vs edge-weighted

- District 2: 55.3% minority (MM)
- Remaining 4 districts: 30-40% minority
- Edge cut: 395 (baseline: 338, +17%)

4.5.4 Weight Factor Sensitivity

Figure ?? shows success rate by weight factor across all states:

- **1x (baseline)**: 40% success (2/5 states) - equivalent to multi-constraint
- **5x**: 60% success (3/5 states) - Alabama breakthrough
- **10x**: 80% success (4/5 states) - Louisiana breakthrough
- **50x-100x**: 80% success maintained
- **500x-1000x**: Success rate decreases (over-weighting causes fragmentation)

Key Finding 6: Moderate weight factors (5x-100x) are optimal. Too low (1x) provides insufficient minority preservation; too high (500x+) causes over-clustering and reduces compactness excessively.

4.5.5 Threshold Sensitivity

Success rate by minority threshold τ :

- **40%**: 65% success (averaged across weight factors)
- **45%**: 70% success
- **50%**: 55% success

- **55%:** 45% success

Lower thresholds (40-45%) perform better because they protect more edges (include "borderline minority" tracts in the protected set), providing algorithm flexibility while preserving core minority communities.

4.5.6 Compactness Tradeoff

Edge weighting increases edge cut compared to baseline, but modestly:

State	Baseline	Optimal Success	Increase	% Increase
Alabama	280	276	-4	-1.4%
Louisiana	338	395	+57	+17%
Mississippi	100	100	0	0%
Georgia	742	742	0	0%
Average			+13	+4%

Table 8: Edge cut: baseline (1x) vs optimal successful configuration

Key Finding 7: Compactness cost is minimal (average +4% edge cut) for achieving VRA compliance. Alabama actually improves compactness (-1.4%) while gaining 2 MM districts.

4.5.7 South Carolina: Remaining Challenge

South Carolina (35.1% minority, target 3 MM) remains challenging even with edge weighting. Best result: 50x @ 50% achieves 49.2% maximum minority (0.8 points short of MM threshold) in 0 districts.

However, this represents progress: multi-constraint maximum was 47.2% (+2.0 percentage point improvement). South Carolina’s low state-wide minority percentage likely represents a fundamental geographic limit even for edge-weighted approaches.

4.5.8 District-Level Compactness Analysis

A critical question: Does creating MM districts through edge weighting sacrifice compactness in non-MM districts? We measure average internal edges per district as a compactness proxy.

Key Finding 8: Edge weighting maintains nearly identical average district compactness (+0.1-0.3% change). This suggests minority concentration does not come at the expense of non-MM district quality.

Mechanism: Edge weighting protects minority-minority edges but leaves other edges at normal weight. METIS still optimizes overall edge cut, so non-MM districts remain compact. Only edges crossing minority community boundaries bear the cost, and these are precisely the edges we want to avoid cutting for VRA compliance.

Future work: Per-district compactness analysis (measuring each district individually) would confirm that MM and non-MM districts maintain similar compactness levels, ensuring the entire districting plan remains balanced.

State	Config	Avg Internal Edges	Change
Alabama	Baseline (1x)	585.1	–
Alabama	Optimal (5x @ 40%)	585.7	+0.1%
Louisiana	Baseline (1x)	226.2	–
Louisiana	Optimal (5x @ 40%)	226.8	+0.3%
Georgia	Baseline (1x)	146.6	–
Georgia	Optimal (1x @ 40%)	146.6	0%
Mississippi	Baseline (1x)	219.5	–
Mississippi	Optimal (1x @ 40%)	219.5	0%

Table 9: Average internal edges per district: baseline vs optimal

5 Discussion

5.1 Ablation vs. Production: Interpreting the 80% vs. 40% Gap

The ablation study (Section ??) achieves 80% success (4/5 states) using per-state-optimal fixed weight factors. The V4 production pipeline uses the adaptive formula (Eq. ??) and achieves full legal targets in 2/5 states (Alabama, Georgia). Understanding this gap is essential for interpreting the paper’s headline claim.

Why the gap exists: The ablation identifies that Louisiana needs $\alpha \geq 5\times$ at $\tau = 0.40$ to achieve 2 MM districts, and Mississippi needs $\alpha \geq 1\times$ (the unmodified graph already achieves the target). The adaptive formula gives Louisiana $\approx 7.9\times$ and Mississippi $\approx 6.7\times$ —well within the ablation’s successful range. However, METIS is stochastic: different random seeds produce different partitions. Mississippi District 1 sits at 49.99% minority in the V4 run verified 2026-04-24—essentially a coin flip. A different run (or the ablation’s run) may produce 2 MM districts.

What 80% means: The ablation result is reproducible from the methods and configurations listed in Section ?. It demonstrates that the edge-weighting approach *can* achieve VRA compliance for 4 of 5 states under optimal per-state parameters. The V4 production formula is designed to approximate those optimal parameters adaptively without state-specific tuning—and succeeds for the two highest-priority states (Alabama, Georgia).

Recommendation: Practitioners seeking to reproduce the 80% headline should use the per-state configurations in Table ?. For production deployment across all states without parameter tuning, the adaptive formula (Eq. ??) provides a reasonable default at the cost of borderline-state reproducibility (Louisiana, Mississippi).

5.2 Geographic Constraints Dominate Algorithm Choice

Our results demonstrate that VRA compliance feasibility depends primarily on geographic distribution of minority populations, not algorithmic sophistication. While n-way partitioning outperforms recursive bisection by 3-7 percentage points, this advantage is insufficient to change outcomes in states with dispersed minority populations.

Alabama illustrates this starkly: After testing six different tree structures, three parti-

tioning methods, and constraint relaxation, the best result (49.6% minority) still falls short of the 50% MM threshold. The 0.4 percentage point gap represents a fundamental geographic limit, not an algorithmic deficiency.

5.3 The VRA-Compactness Tradeoff

VRA compliance and compactness optimization are fundamentally in tension:

VRA requires: Concentrated minority populations in specific districts (non-uniform distribution).

Compactness requires: Minimizing edge cuts, which favors uniform population distribution.

METIS edge-cut minimization naturally produces districts with similar demographic profiles. Creating MM districts at 60% minority in a 37% minority state requires deliberately non-compact boundaries to reach sufficient minority population.

Our 42% threshold reflects the point where these goals become compatible: States above this threshold can achieve VRA compliance while maintaining compact districts. Below this threshold, one goal must be sacrificed.

5.4 Policy Implications

5.4.1 Feasibility Assessment

Jurisdictions can estimate VRA compliance feasibility before detailed redistricting:

$$\text{Feasible} \iff m_{\text{state}} \gtrsim 0.42 \text{ and clustering favorable} \quad (10)$$

For borderline states (40-42%), success depends on geographic clustering patterns. Louisiana (41.6%) achieves 1 of 2 MM districts - might achieve 2 with less emphasis on compactness.

5.4.2 Algorithmic vs. Political Redistricting

Our findings suggest limits of purely algorithmic approaches for VRA compliance:

Where algorithms work (GA, MS): High minority percentage + favorable geography enable VRA compliance with compact districts. Algorithmic methods appropriate.

Where algorithms struggle (AL, SC): Low minority percentage + dispersed geography require tradeoffs. Current practice of political map-drawing may be necessary, but creates opportunities for partisan manipulation (?).

5.4.3 Legal Considerations

Courts must balance competing redistricting principles (?). Our results inform this balance:

- **If $m \geq 0.42$:** VRA compliance achievable with compact districts. No excuse for failure.
- **If $0.40 \leq m < 0.42$:** Tradeoffs necessary. Slight compactness reduction may enable compliance.

- If $m < 0.37$: VRA compliance may be infeasible without substantial compactness sacrifice. Alternative remedies (proportional representation, at-large elections with cumulative voting) merit consideration.

5.5 Comparison to Prior Work

5.5.1 MCMC Approaches

Duchin et al. (?) analyze Alabama through MCMC ensembles, finding difficulty creating 2 MM districts. Our results align: Alabama’s geography fundamentally limits VRA compliance. However, our multi-constraint optimization approach provides:

- Deterministic results (vs. stochastic sampling)
- Explicit target specification
- Clear feasibility thresholds

5.5.2 Single-State Studies

Chen (?) examines Texas (42.6% minority) and finds VRA compliance achievable. This aligns with our 42% threshold - Texas sits just above it. Our multi-state comparison provides broader context than single-state analyses.

5.5.3 Optimization Approaches

Integer programming methods (?) can explicitly encode VRA constraints but scale poorly (> 1000 tracts). METIS handles large graphs efficiently while producing reasonable solutions, though not provably optimal.

5.6 Methodological Insights

5.6.1 Why N-Way Outperforms Recursive Bisection

N-way partitioning considers all edge cuts simultaneously, enabling global optimization. Recursive bisection makes greedy decisions at each split, potentially missing better configurations.

For Alabama, n-way achieves 47.3% vs. recursive’s 43.0%. The 4.3 percentage point difference reflects n-way’s ability to consider the full problem structure rather than intermediate constraints.

5.6.2 Adaptive Bisection Improvements

Adaptive bisection’s 3.1 percentage point gain over predetermined trees (46.1% vs. 43.0%) shows value of data-driven split decisions. However, adaptive remains bounded by recursive framework’s greedy nature - cannot match n-way’s global view.

5.6.3 Diminishing Returns of Constraint Relaxation

ubvec provides 2-3 percentage points in difficult states but hits geographic ceiling. Alabama’s 49.6% with ubvec=1000 suggests further relaxation (ubvec=10000+) would yield minimal additional gain.

5.7 Limitations and Future Work

5.7.1 Compactness Metrics

We use edge-cut minimization as compactness proxy. Alternative metrics (Polsby-Popper, Reock) might yield different results. However, Alabama’s 49.6% ceiling suggests geography dominates metric choice.

5.7.2 Target MM Percentage

We target 60% minority for MM districts. Reducing to 55% or 52% might enable more states to meet VRA requirements, at cost of less secure MM districts. Future work could explore this tradeoff systematically.

5.7.3 Block-Level Data

Census block data (higher resolution than tracts) might enable more precise minority concentration. However, computational cost increases substantially (10× more vertices). Preliminary tests show minimal improvement - geography remains limiting factor.

5.7.4 Multi-Objective Optimization

Explicit multi-objective optimization (Pareto frontier of VRA vs. compactness) could quantify tradeoffs more precisely. Current work uses implicit tradeoff through constraint relaxation.

5.7.5 Additional States

Testing additional VRA-covered states (Texas, North Carolina, Florida) would strengthen threshold estimates. Texas (42.6%) particularly interesting as borderline case.

5.8 Edge-Weighting: A Breakthrough Approach

Multi-constraint optimization (Methods 1-3) achieved limited success (40%), but edge-weighted single-objective optimization (Method 4) doubled this rate to 80%. Three factors explain this breakthrough:

5.8.1 Direct Encoding of VRA Goals

Multi-constraint treats VRA as an optimization constraint: balance population and minority VAP simultaneously. This is indirect—METIS must discover which tracts to group to achieve target minority percentages.

Edge-weighting directly encodes the VRA principle: *"don't split minority communities"*. Edges between minority tracts receive high weights, making them expensive to cut. METIS's edge-cut minimization naturally preserves these communities without explicit minority balancing.

This is conceptually cleaner: rather than telling the algorithm *what outcome we want* (60% minority districts), we tell it *which structures to preserve* (minority community boundaries).

5.8.2 Single-Objective Simplicity

Multi-constraint optimization requires balancing competing objectives. When population balance demands one split and minority concentration demands another, METIS must compromise. The ubvec parameter (allowing $1000\times$ minority imbalance) attempts to resolve this but fundamentally maintains the dual-objective framework.

Edge-weighting uses single-objective optimization (minimize weighted edge cut) with only population balance as a constraint. This simplifies the optimization landscape, potentially enabling better solutions.

5.8.3 Alabama as Critical Case

Alabama's 49.6% ceiling with multi-constraint (0.4 points short of 50%) appeared to be a fundamental limit. Edge-weighting crosses this threshold, achieving 50.8% and creating 2 MM districts.

The key: identifying *which* 0.4% of minority population to concentrate. Multi-constraint spreads minority population more uniformly across districts. Edge-weighting preserves concentrated minority clusters by protecting their internal boundaries, enabling the marginal concentration needed for VRA compliance.

5.8.4 Compactness Preservation

Surprisingly, edge-weighting maintains similar compactness to baseline (average +4% edge cut, Alabama actually -1.4%). Why?

Mechanism: Only edges between minority tracts receive high weights. METIS still optimizes overall edge cut, so it maintains compact districts where possible. The only edges bearing higher cost are minority community boundaries—precisely the edges VRA principles suggest we should avoid cutting.

Non-MM districts remain compact because their internal edges have normal weights. The compactness "cost" is localized to minority district boundaries, not distributed across the entire state.

5.9 Broader Implications

Edge-weighting resolves a tension previously thought fundamental: achieving VRA compliance without sacrificing algorithmic principles or compactness. The approach:

- **Maintains algorithmic transparency:** Single-objective optimization with explicit, reproducible parameters
- **Achieves minority representation:** 80% success rate (4/5 states) vs 40% for multi-constraint
- **Preserves compactness:** Minimal edge cut increase (average +4%)

This suggests limits of algorithmic redistricting are *methodological*, not fundamental. Prior approaches (multi-constraint) couldn't achieve VRA compliance in dispersed-minority states. Edge-weighting overcomes this limitation by better encoding the underlying principle.

5.9.1 When Edge-Weighting Fails

South Carolina (35.1% minority) remains challenging even with edge-weighting (maximum 49.2%, 0.8 points short). This likely represents a true geographic limit: state-wide minority percentage is too low to concentrate into 3 MM districts at 50%+ without extreme compactness sacrifice.

For such states, alternatives remain necessary:

1. **Accept tradeoffs:** Prioritize VRA compliance over strict compactness
2. **Alternative systems:** Proportional representation, ranked-choice voting, multi-member districts
3. **Hybrid approaches:** Algorithmic baseline with targeted adjustments

But edge-weighting *narrows* the set of states requiring these alternatives from 60% (3/5: AL, LA, SC) to 20% (1/5: SC only).

6 Conclusion

We systematically evaluated VRA compliance through principled optimization methods, testing four approaches across five covered states. Our investigation yields four primary conclusions:

(1) **Multi-Constraint Limitations:** Multi-constraint optimization (n-way partitioning, adaptive bisection, constraint relaxation) achieved only 40% success (2/5 states). Advanced methods improved concentration by 3-7 percentage points but could not overcome geographic constraints.

(2) **Edge-Weighting Breakthrough:** Single-objective optimization with weighted edges doubles success rate to 80% (4/5 states) with per-state-optimal configurations, versus 40% for multi-constraint. With the V4 production adaptive formula (Eq. ??), full legal

targets are met in 2/5 states; Louisiana and Mississippi each achieve 1 of 2 target districts due to stochastic boundary sensitivity (see Section ??). Alabama breakthrough: 0 MM $\rightarrow$ 2 MM districts (50.8% minority, confirmed in production).

(3) Minimal Compactness Cost: Edge-weighting increases average edge cut by only +4% compared to baseline. Alabama actually improves compactness (-1.4%) while gaining 2 MM districts. This resolves the perceived tension between VRA compliance and algorithmic principles.

(4) Revised Feasibility Threshold: With edge-weighting, states with $\geq 37\%$ minority can achieve VRA compliance (AL: 36.9% succeeds). Only states with $< 35\%$ minority remain challenging (SC: 35.1% reaches 49.2%, 0.8 points short). The critical threshold shifts downward from 42% to 36%.

6.1 Contributions

This work makes four key contributions to redistricting science and VRA compliance:

(1) Methodological Innovation: Novel edge-weighted single-objective optimization approach that doubles VRA success rate (40% $\rightarrow$ 80%) compared to multi-constraint methods. Direct encoding of community preservation principle into graph structure.

(2) Systematic Evaluation: First comprehensive comparison of four optimization approaches (recursive bisection, n-way, adaptive, edge-weighted) across multiple VRA-covered states. Ablation study of 140 configurations (7 weight factors $\times$ 4 thresholds $\times$ 5 states) quantifies parameter sensitivity.

(3) Empirical Thresholds: Identification of revised feasibility threshold (36

(4) Policy Framework: Clear methodology for assessing VRA compliance feasibility and selecting appropriate optimization approach based on state demographics. Demonstrates algorithmic redistricting can achieve VRA compliance in most covered jurisdictions.

6.2 Recommendations

For practitioners and policymakers:

Jurisdictions with $m \geq 0.37$: Use edge-weighted single-objective optimization. Start with weight factor 5x-10x at 40-45% threshold. VRA compliance achievable with minimal compactness cost (+4% average edge cut). Algorithmic methods fully appropriate.

Jurisdictions with $0.35 \leq m < 0.37$: Edge-weighting may approach but not achieve VRA compliance. Test with higher weight factors (50x-100x) and lower thresholds (40%). If unsuccessful, consider:

- Hybrid approach: algorithmic baseline with targeted adjustments
- Modest compactness reduction for VRA priority
- Alternative electoral systems (proportional representation, multi-member districts)

Jurisdictions with $m < 0.35$: Algorithmic single-member districts likely insufficient. Alternative remedies necessary: proportional representation, multi-member districts with ranked-choice voting, or cumulative voting.

Courts: Use 36

Parameter Selection:

- **Weight factor:** Start low (5x-10x) for principled approach. Increase to 50x-100x only if necessary. Avoid 500x+ (causes over-clustering).
- **Threshold:** Use 40-45% for borderline states (includes "near-minority" tracts), 50% for high-minority states.
- **Validation:** Measure both VRA compliance and compactness. Accept configurations with $\leq 10\%$ edge cut increase.

6.3 Future Directions

This work opens several research directions:

(1) **Full National Analysis:** Extend to all 50 states with complete demographic data. Characterize feasibility across full range of minority percentages (10-50%).

(2) **N-Way vs. Recursive Bisection Theory:** Formal analysis of when n-way and recursive bisection produce equivalent results. Our empirical work shows differences, but theoretical characterization remains open.

(3) **Block-Level Analysis:** Rerun analysis with census block data ($10\times$ resolution). Determine whether finer granularity enables better minority concentration or whether geographic patterns dominate at all scales.

(4) **Multi-Objective Optimization:** Explicit Pareto frontier analysis of VRA compliance vs. compactness. Quantify exact tradeoff curves rather than binary success/failure.

(5) **Alternative VRA Targets:** Explore 52%, 55%, 65% MM district thresholds. Determine how threshold choice affects feasibility and representation security.

(6) **Temporal Analysis:** Apply methods to 2010 and 2000 Census data. Examine how changing demographics affect VRA compliance feasibility over time.

(7) **Alternative Electoral Systems:** Design proportional representation or ranked-choice systems that achieve minority representation without geographic concentration constraints. Compare representation quality to single-member districts.

6.4 Closing Remarks

The Voting Rights Act represents a crucial protection against minority vote dilution, but our results demonstrate fundamental limits to achieving VRA compliance through algorithmic redistricting in states with dispersed minority populations. In these cases, the choice is not between gerrymandering and principled methods, but between:

- **Prioritizing compactness** at the expense of minority representation
- **Prioritizing VRA compliance** at the expense of compactness
- **Adopting alternative systems** that avoid the geographic concentration problem

Our work provides the quantitative foundation for making this choice informed by data rather than rhetoric. For states above the 42% threshold, there is no excuse for VRA non-compliance - algorithmic methods prove compliance is geometrically feasible. For states below 37%, policymakers must confront the inherent tension between competing principles.

As redistricting reform efforts nationwide emphasize independent commissions and objective criteria (?), our findings highlight the importance of geographic and demographic context. One-size-fits-all approaches may be insufficient. Instead, redistricting standards should be calibrated to jurisdictional demographics, with VRA compliance feasibility assessments informing appropriate criteria weights.

Ultimately, achieving both compact districts and minority representation may require rethinking the single-member geographic district paradigm itself. Multi-member districts, proportional representation, or ranked-choice voting systems can provide minority representation without the geographic constraints that limit algorithmic redistricting. Our work quantifies exactly when these alternatives merit serious consideration.

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